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ORIGINAL PAPER

Recent progress in sign language recognition: a review

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Abstract
Sign language is a predominant form of communication among a large group of society. The nature of sign languages is visual, making them distinct from spoken languages. Unfortunately, very few able people can understand sign language making communication with the hearing-impaired infeasible. Research in the field of sign language recognition (SLR) can help reduce the barrier between deaf and able people. Despite having tremendous advances in SLR, unfortunately, this form of recognition is still at least a decade behind speech recognition. There has been a gradual transition from static to isolated to continuous SLR, but still the research is scattered, limited to very small vocabularies, and only suitable for tailor-made conditions. This paper aims to compile recent progress in SLR and presents a comprehensive review of the emerging SLR frameworks and algorithms. We have categorized SLR based on the unit of written text, i.e., letters or alphabets, words and sentences. This review also includes a study-wise summary of the datasets used in different research conducted during the last few years. We identify state-of-the-art techniques for each category. We also suggest novel research directions for future work, and highlight several primary factors contributing to SLR's inability to achieve improved practical outcomes.

Keywords Sign language recognition · Transformer · Gesture recognition · Continuous SLR · Isolated SLR

1 Introduction

Sign language is a basic form of communication among the population with hearing disabilities. There is more than 5% percent of the population that faces hearing disabilities that make up around 360 million people in the world [1]. Additionally, over 70% of these individuals use sign language to convey their thoughts and emotions, out of which 80% belong to developing countries [2]. They use about 300 different sign languages for communication. Sign language helps these specific communities to learn, connect with others in the community, work and even be able to retrieve services.

Sign language is a fusion of manual and non-manual representations of signs used by these deaf people [3]. Manual signs consist of movements of arms and hands while the non-manual form contains posture and facial expressions. Sign languages consist of grammar and rules, as any other natural language does. Moreover, sign languages are not universal and have been developed for specific languages such as American Sign Language (ASL), Chinese Sign Language (CSL), Pakistan Sign Language (PSL), etc.

Sign Language is extremely vital for growing children with hearing disabilities. Many researchers have found that sign language helps in speech development which enables young children to convey their emotions easily thus decreasing frustration in them. The ability to 'express' these emotions lead to a healthy parent–child bond. Furthermore, this helps in growing the intelligence of the child. It further enables early brain growth and even quick motor-skill development. Sign language may also assist deaf people to excel in the professional world. Furthermore, improved communication enables individuals to enhance their career growth by acquiring skills more easily. Moreover, job interviews can be easily tackled with the use of sign language for these individuals.

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To be able to effectively communicate with individuals who use sign languages, research on performing automated SLR using artificial intelligence is conducted by experts in computer science. Currently, there is active research on developing deep learning-based models to recognize sign languages for various languages covering various domains. However, there are several challenges such as, recognition of hand poses and gestures, signs having similar appearances etc which increases the complexity and computational cost of the models. Many studies focus on the recognition of single alphabets of the language as these are the basis of sign language and then construct to words and sentences. Simple to complex systems are developed, trained, and tested on various datasets in order to overcome the challenges and improve the accuracy of the recognition.

It is noted that sign languages are based on various datasets. The most commonly used dataset is ASL which is easily available online and representations are even available in the form of videos on Youtube. Another dataset is CSL [4] which, as the name suggests, is based on the Chinese language. German sign language (GSL) was also used in some papers but these papers were based on Germany due to the complexity of the German language itself. The source of this dataset is mainly a German weather forecasting channel. One paper was also seen on the Greek Sign Language [5] which tells that work on sign languages is being done in Greece as well.

In this paper, we present a comprehensive review of the studies on SLR which is extremely necessary to end the communication barrier between able and hearing-impaired individuals who can only communicate through signs. We focus on research from the years 2019–2022. The contributions of the paper are threefolds. Firstly, we have categorized SLR based on the unit of written text: letters or alphabets, words and sentences. There have been a few surveys on SLR but their focus is different. For example, [6] and [7] only focus on the deep learning model and have a very limited discussion on sentence-level sign recognition. Another review [8] does not include word-level or isolated studies and only covers studies before 2021. While [9] is a very brief review covering 20 papers. Secondly, this study provides a detailed list of datasets that have been used for SLR from 2019 to 2022. Thus, providing the resources in one place that are required for conducting fruitful research. Thirdly, this paper provides open research questions and future directions in SLR.

The rest of the paper is organized as follows. Section 2 gives a detailed review of SLR techniques based on the proposed taxonomy. Section 3 highlights the data sets used in recent studies. Open research questions and research directions are suggested in Sect. 4 and conclusions follow in Sect. 5.

2 Taxonomy of sign language recognition

At a high-level, we have categorized SLR on the basis of the unit of written text: letters or alphabets, words and sentences. The complete taxonomy of SLR is shown in Fig. 1. Sign words or isolated signs are sequences of hand or gestures where the entire gesture maps to exactly one word. Sign words can also be static images where one image maps to one word. A sign sentence, also called a continuous sign, consists of more than one word.

2.1 Alphabets

Sign alphabets or static signs are fixed images of the shape of a hand. Some of the signed letters from Arabic alphabets are shown in Fig. 2. Sign alphabet recognition is one of the least complex forms of recognition and in this section, we explore various studies that focus on sign alphabet recognition.

2.1.1 Convolutional neural network

We begin with a study [11] that develops a model for skin tone identification in different lighting conditions to improve the accuracy of the recognition systems. This paper also improved the edge recognition capabilities to more accurately detect hand movements. A very basic deep learning model called convolutional neural network (CNN) was used in this study. The *CNN* model consisted of two main stages: feature extraction, to extract important features from an image. This is done by passing the image through a series of convolutional filters and pooling layers. In the second stage, the features are fed to a fully connected layer for training. This layer has a two-layer neural network. Once trained, the model takes in an image and outputs the class label for the sign in the form of text and audio. Deep learning has been used for both text [12] and images [13, 14]. Evaluation metrics used for analysis were recall, precision, f-score, and specificity The highest precision reached by this model was 90% in a real-time detection setting. It would be worth employing more state-of-the-art CNNs such as densenet121 or VGG-16.

Similarly, [15] also used a CNN model called SLRNet-8 to improve the performance of ASL recognition. SLRNet-8 is just another architecture of CNN that consists of 6 convolutional and 3 pooling layers. To validate the model, four datasets were used consisting of the alphabets and numerals of four publicly available ASL datasets. SLRNet-8 outperformed existing models by a margin of 9%. Researchers rely on synthetically generating more data instead of collecting new ones. This is called data augmentation. Data augmentation has been used successfully in various applications such as [16–19]. In this paper, data augmentation techniques were also used to increase the number of training examples such as change of angles from −10 to 10 degrees, zoom 10%, etc.

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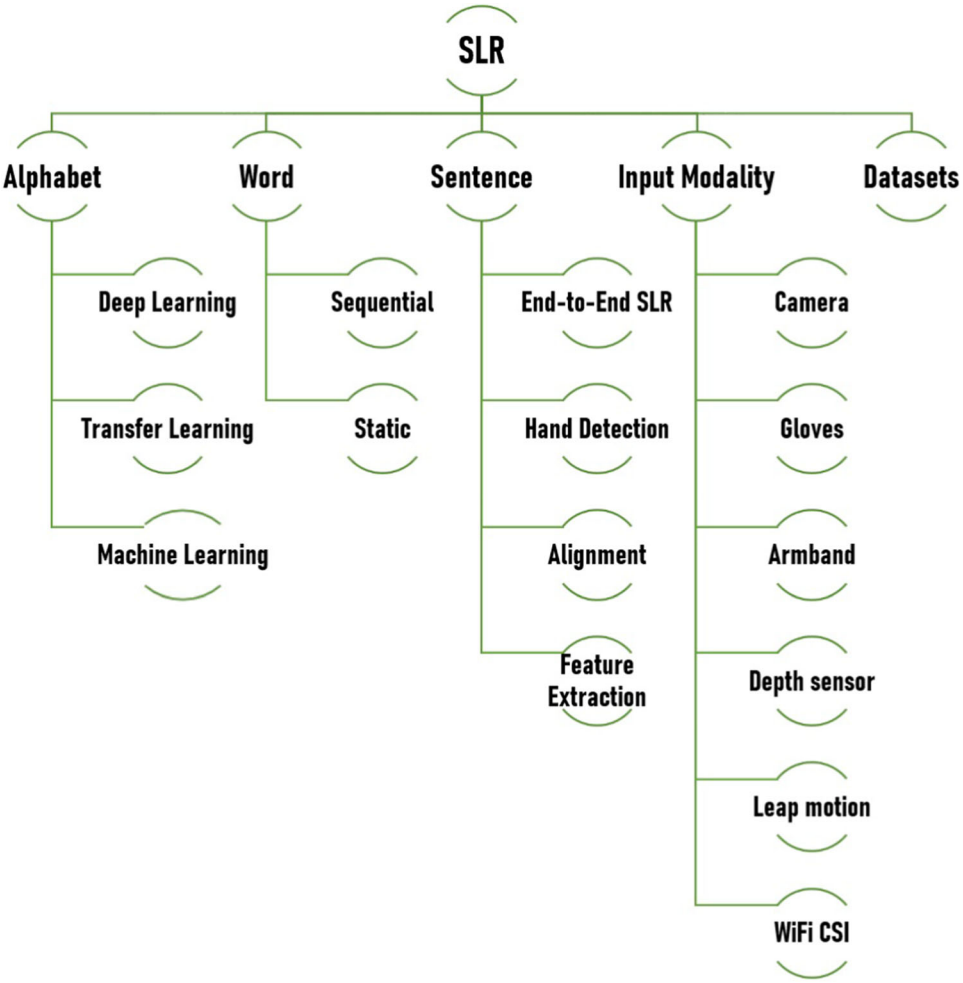


Fig. 1 Taxonomy of sign language recognition

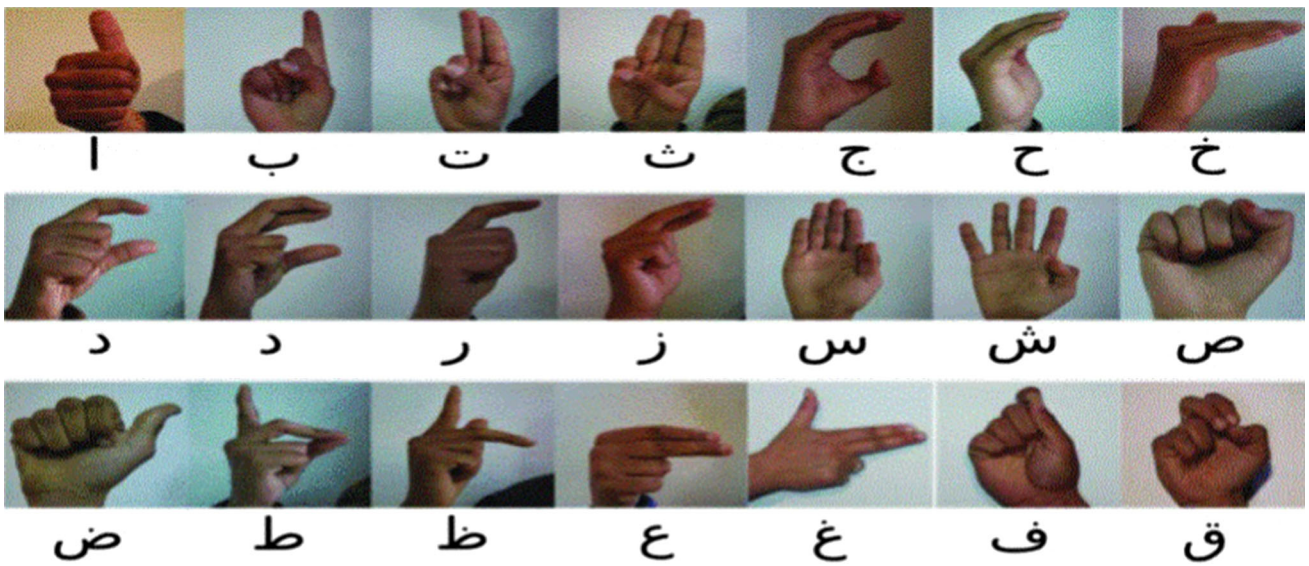


Fig. 2 Static sign images of some Arabic alphabets [10]

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Mannan et al. [20] also used their deep architecture of CNN for the recognition of the 24 English alphabets excluding J and Z. Deep models require a lot of training data. For this purpose, data augmentation methods were also used in this study. The images were scaled and rotated to create variations and thus avoid the detection of unnecessary features. The deep CNN architecture consisted of three convolutional layers to avoid overfitting. The experiment was performed six times with different learning rates. The model was able to detect signs with an accuracy of 99.67%. It was also suggested that the proposed model be extended for real-time recognition in the future through video frames and leap motion controllers. Another study that also used CNNs is [10], but the sign language is Arabic. Specifically, LeNet-5 architecture was used yielding an accuracy of 90.02% thus outperforming k-nearest neighbors (KNN) and support vector machines (SVM) classifiers. Inception-Net, another variation of CNN, was used in [21] for ASL alphabets. Finally, [22] used deep CNN for 10 alphabets of ASL, and produced an accuracy of 99%.

A more mentionable contribution in SLR using CNN is [23] in which a mechanism to translate gestures and hand poses of sign language to natural language is devised. The dataset established in this article includes ASL alphabets (ASLA) developed under varying lighting and background conditions which is actually the main contribution of this study. The dataset on static sign gestures was captured through laptops and smartphones. This new dataset with its own challenges in the form of varying environments raises interesting research questions. Additionally, two other datasets were included in this study to validate the CNN model. The model takes in images of dimensions 64 x 64 x 3. There are 3 convolutional layers. The ReLu operation is performed after each convolution layer. The interface developed in this study for real-time recognition of ASLA employs the OpenCV library to capture live videos that are then split into images. The application then gets color images of dimensions 64 x 64 which are fed to the CNN model for training and classification. The reported accuracy on ASLA datasets was 99.41%. This study also compared the proposed architecture with other architectures of CNN and SVM.

2.1.2 Transfer learning

In another study on CNN [24], a more advanced EfficientNetB4 model for Arabic SLR (ArSL) is proposed. The dataset consisted of 32 Arabic alphabets. The proposed model employed transfer learning. In *transfer learning*, a CNN already pre-trained on some other dataset is used. From this trained model, the feature extraction part is separated from the fully connected layer. The feature extraction part is retained and used to extract features of the Arabic dataset. These features are then used to train a fully connected layer from scratch. Thus, a network that was trained on one dataset is used (transferred) to extract features of another one. For this study, the pre-trained model was EfficientNetB4. Based on the descriptors of the dataset: the number of features and classes or alphabets, the model is modified. The ArSL dataset was first augmented to increase the training data. For this purpose, an ImageDataGenerator provided by the Keras API was used. The purpose of increasing the dataset is to avoid overfitting. Precision, recall, and F1-Score was the evaluation metrics used for this model. Testing accuracy of 95% was achieved which was higher than some of the previous models mentioned in the article which only had 20 classes while ArSL used in this study had 32 classes.

Another study, [25], also employed transfer learning using a pre-trained VGG-16 model. As the name suggests, VGG-16 has 16 layers and is yet another architecture of CNN. This model was used for ASL. In this study, two-way communication is enabled by including a text-to-sign conversion mechanism. One of the main contributions of this study is the dataset, that was created and introduced. This dataset had letters A to H. The images were gathered by a webcam using a data collection program. A total of 2500 images were collected. Like [24], data augmentation was also performed in this study such as horizontal flipping, gamma and Gaussian noise. The pr-trained VGG-16 was fed these images for feature extraction. Some modifications had to be done to VGG-16 to make it dataset complaint. The modifications included removing the top layer so the model would accept an image of dimensions (50,50,3) and changing the number of nodes in the output layer to 8, one for each letter. The model was trained for 20 epochs and testing accuracy of 99.62% was achieved. Transfer learning was also used in [26] for Turkish sign language (TSL) recognition. In this case, pre-trained AlexNet was used. Region-based CNN (R-CNN) was used as a classifier to learn these features.

2.1.3 Machine learning and other techniques

Moving on to more basic models, [27] used several classification techniques such as HMM, HSBN, BoostMap, and neural networks for sign language-related gestures. Both static and dynamic datasets were collected, however, dynamic datasets were more challenging and hence the research was narrowed down to only the static domain. The dataset was in the form of images (or videos) taken from a digital camera. In addition, a variety of lighting settings along with different variations such as caps, glasses, etc. were considered. The approach is to determine the shape of the hands and their center of gravity from which the distance of each finger is calculated to determine the exact angle and shape of the gesture. These angles and shapes, also called landmarks, are then used to train the classifiers. The images are converted to binary to speed up classification.

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A similar type of hand landmarks were also used by [28] as features. It mainly focuses on the recognition of ASL and Irish sign language (ISL-HS). This paper uses the mediapipe implementation available in OpenCV. *Mediapipe* is the library used for computer vision analysis. It first detects the hands and feeds them to a landmark detection model. This model extracts 21 marks for each hand. Like[27], lines and angles are extracted which are used as the input for classifiers. The high-level diagram of the model is given in Fig. 3. Five different classifiers were used in this study: naive Bayes, decision trees, random forest, k-nearest neighbors (KNN), and SVM. It was shown that the random forest classifier gave the best results with the proposed model whereas the na–ve Bayes classifier performed poorly. The reason for low accuracy could be the inter-dependency of features (lines and angles) as the na–ve Bayes classifier works better with independent features. The top accuracy obtained with ASL and ISL-HS datasets were 93% and 96.7%, respectively.

Mediapipe was also used by [29] for extracting 21 landmarks for ASL recognition. The article experimented with 28 different landmark combinations and utilized the KNN, random forests, and neural networks as classification algorithms. The main purpose of the article is to bring forth the best pre-processing and algorithm pair. Once the landmarks are obtained, pre-processing is performed to bring them to a single reference frame since all these coordinates can be anywhere in the image frame. The pre-processing involved the use of a bounding box for calculations and then normalization. The types of bounding boxes used were cuboidal and cubical. Bounding boxes bounded the 21 coordinates of the hand gestures. Furthermore, the mathematical transformations employed to normalize coordinates into a common reference frame were shifting and scaling. The KNN algorithm had an average accuracy of 82.19%. The random forest had an average accuracy of 85.30% while the neural network had an average accuracy of 90.95%.

In [30], multiple kernel learning classification in SVM was used to recognize 36 static alphabets of Pakistani Sign Language. Images from PSL videos are collected and then segmented using K-mean clustering after being converted to grayscale. Then four vision-based features: local binary patterns (LBP), a histogram of oriented gradients (HOG), an edge-oriented histogram (EOH), and speeded-up robust features (SURF) are extracted individually. The classification of these features is done through multiple kernels as suggested in the paper, namely Gaussian, linear and polynomial kernels. After extraction of these four features, they are classified using kernels. The selection of the best kernel is done through an algorithm named Kernel Selection Algorithm (KSA). The KSA takes in extracted features and uses k-fold validation with k=10 to calculate the highest average accuracy for each feature. The kernel with the highest accuracy for a specific feature space is selected individually. The output of these kernel models is ensembled using the Ensemble of Feature-Based Kernels Algorithm (EoFKA) algorithm to give the final predicted output class. The features linear for HOG had the highest average accuracy of 89.24%.

2.1.4 Summary and conclusion

Signed letter recognition is similar to problems like character, shape or object recognition. Based on the review, the suggested method for performing alphabet-based SLR recognition is through a pre-trained VGG-16 using transfer learning. A summary of all studies covered in this section is given in Table 1. Only the top accuracy reported in the study is shown in this table.

2.2 Words

In this section, we cover word-level SLR. In this type, the prediction made by the SLR systems is that of a word. Word-level SLR can further be categorized as sequential or static. In sequential mode, the word is represented as a sequence of hand and arm movements. In static mode, a word is represented by an fixed shape of the hands. We begin the discussion with sequential mode.

In [32] a high-level ASL classifier for a word-based dataset was presented. The classifier makes use of a holistic-based approach and a 2D human posed-based approach. The first baseline model employs transfer learning to extract spatial features from the given videos. For this purpose VGG16 pre-trained on ImageNet was used. Only 50 frames were randomly selected from each video. The extracted features were passed onto the gated recurrent unit (GRU). For the pose-based model, 50 body and hand points are selected using OpenPose and then fed to GRU. The paper also proposed a temporal graph convolution network (TGCN) as a pose-based baseline. Here, the human body was considered a fully connected graph, and a weighted adjacency matrix is created accordingly. This graph was passed to the neural network and was trained according to the equation:

$$H_{n+1} = Gn(H_n) = \sigma(A_n H_n W_n) \tag{1}$$

,

Both VGG-16 and TGCN performed equally with an accuracy of around 62% on 2000 gloss words.

Similarly, [33] worked on a word-based dataset to emphasize the context as well as the gesture while passing it into a training model. It was noted that although the dominant hand tends to carry significant information regarding the meaning of a sign, other features such as facial expressions and the surroundings also play a key role in identifying the correct meaning. According to the proposed pipeline, spatial features are extracted from the data frame through the 2DCNNs

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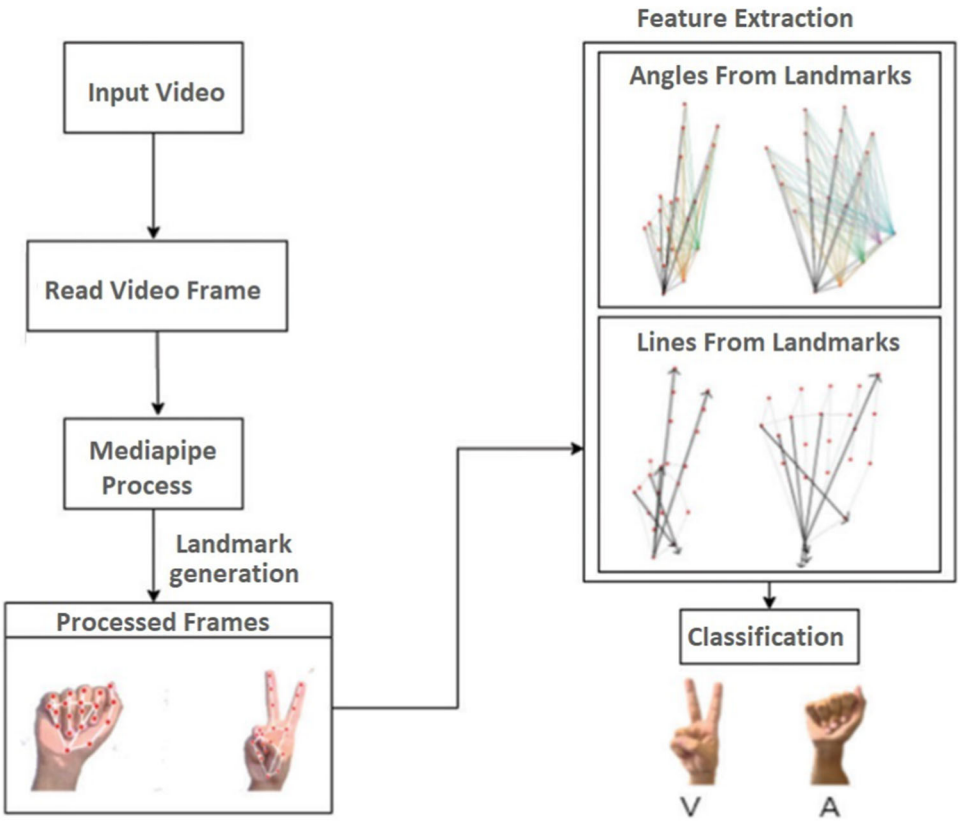


Fig. 3 High-level diagram explaining how angles and lines were extracted from landmarks in the alphabet-level SLR system given in [28]

Table 1 A summary of sign recognition studies on static signs representing alphabets

References	Model/technique	Dataset	Best acc
[10]	Le-Net5	ArSL	90.02%
[11]	CNN	Italian gestures	90%
[15]	CNN	ASL	Up by 9%
[23]	CNN	ASLA	99.41%
[22]	Deep CNN	ASL (10 alphabets)	99.0%
[28]	NB, DT, RF, KNN, SVM. 21 landmarks	ASL, Irish SL	93%, 96.7%
[20]	Deep CNN + Augment	ASL (excluding J and Z)	99.67%
[24]	CNN (transfer) + Augment	ArSL (32 alphabets)	95%
[25]	VGG-16 (transfer)	ASL (A-H)	99.62%
[26]	AlexNet (transfer)	TSL	99.7% (prec.)
[27]	HMM, ANN	–	–
[29]	KNN, RF, ANN + Augment. Hand track points	ASL	90.95% (KNN)
[30]	SVM. Feature extraction	PSL (36 alphabets)	91.98%
[31]	SVM, CNN, SURF	ISL (alphabets. and digits)	99.0%

TSL, Turkish SL; ArSL, Arabic SL; PSL, Pakistan SL

model MobileNetV2 which has low latency and high efficiency as a feature extractor. Then the model uses temporal modeling to understand temporal changes before learning the sequence alignment using the connectionist temporal classification (CTC).

Before proceeding further, *CTC* is briefly discussed. The timing of a sign varies from person to person which makes it difficult to come up with a basic mapping such as x frames map to one gloss. As a result, aligning multiple signs in a video with glosses is a challenge. In 2006, Graves et al. proposed CTC to align elements in one sequence with the elements in another sequence [34]. This technique is therefore more useful in sentence-level SLR, but can also be used for isolated SLR. The basic idea of CTC is to take a fragment

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of a signed video at every time step t and assign a label (gloss) to it based on probability. Multiple consecutive ts may get the same label, so it uses collapsing to reduce repeating glosses to one. CTC also takes care of repeating letters in words like 'food'. CTC utilizes a loss function based on the probabilities of all time steps. Its goal is to find a t that minimizes the loss.

Coming back to [33], it is also important to note that the model is trained on two streams, one is the context stream which contains full information, while the other is the hand stream which focuses on the dominant hand only. A Layer-Norm operation is applied to the hand stream and the information is added to the original stream.

In [35], the authors talk about selecting relevant sign-language data from a stream of abundant sources to fine-tune the model. This paper also demonstrates how effectively transfer learning can be applied to isolated SLR using an inflated 3D (I3D) CNN as the deep learning architecture. The model is trained on a single Nvidia Titan RTX graphics card utilizing the CUDA platform. Here the model is first pre-trained on the MS-ASL dataset and then it was fine-tuned on SIGNUM to implement the GSL. Both of these architectures made use of Inception-v3 2D convNet with an input image size of 224x224 pixels where the temporal depth of 20 reduced to 6 by the last pooling layer. ImageNet was also used to extract spatial features from the frame. The performance of the model went up by 21% as compared to models that were not pre-trained on MS-ASL datasets.

In the same way, [36] presented a model for gesture-based SLR known as the G-CNN. The study also compares the performance of the proposed model with the VGG-11 and VGG-16 models. Several experiments are carried out to evaluate the efficiency of the three aforementioned models. The VGG-11 and VGG-16 models are modified to cater to the prepared dataset. The experiments performed on the ISL dataset show the highest classification accuracy for G-CNN for both original and augmented data at 94.83% and 96.43%, respectively. Moreover, G-CNN converged faster than the other two models.

In [37], hand-tracking and hand representation models are presented that can later be used for ASL recognition. Hand-tracking is done using a particle filter. A CNN model pre-trained on hand dataset is used to verify the presence of the hand in the regions identified by the dynamic model of the particle filter. Based on the output of the particle filter and pre-trained CNN model, the hand regions are extracted and given to the hand representation model. This model simply takes the mean of all hand regions resulting in a hand energy image (HEI). A few examples that show how HEI is calculated are given in Fig. 4. The classification algorithm simply calculates the distance of the HEI of test sample with all training samples. The proposed model showed the best performance when compared with other hand-tracking methods. The model was evaluated in two isolated ASL datasets RWTH-BOSTON-50 and ASL dataset. These datasets consisted of indoor videos.

Furthermore, [38] proposed an isolated (word-level) SLR model based on motion history images (MHI) that are grayscale static images that record the movement and path of the hands (or objects). In this study, RGB-MHI are created from RGB video frames. These images represent a spatio-temporal summary of a sign as a single RGB image. The initial step was to create an RGB-MHI images from sign videos. Two SLR models were proposed. In the first model, In the first one, the RGB-MHI model is used as an attention medium in a 3DCNN (I3D) to focus on the important parts of the sign video. The second model employs the late fusion technique while using the RGB-MHI model features (RGB-MHI images and RGB-MHI frames together) directly with the 3DCNN model features. The RGB-MHI model was constructed using ResNet50 pre-trained on the ImageNet dataset. However, changes were made to the ResNet layers by removing the global average pooling (GAP) layer and adding an extra fully connected layer with 1024 neurons that had ReLu operations. As a pre-processing step, a pre-trained faster R-CNN model along with ResNet-50 is used to detect the signer and remove unnecessary background noise. Data augmentation was also performed to extend the datasets. The results of the experiments showed that the best results were achieved when all the layers of ResNet-50 were used on the AUTSL dataset in the RGB-MHI model with an accuracy of 71.95%. An accuracy of 89.94% was achieved when the self-attention model with the I3D model was used. The RGB-MHI attention model used with the I3D showed the highest accuracy of 90.18%. The accuracy further increased to 93.53% with data augmentation.

Similarly, a CNN-based approach was employed by Venugopalan et al. [39]. This article used a unique approach of combining two different types of classifiers to detect Indian Sign Language (ISL) gestures related to some common words used by COVID-19 patients in emergencies. The proposed model consisted of two different deep neural networks combined. The first one was a CNN-based feature extraction model known as VGG-16 which was responsible for feature extraction. A sequence of feature vectors extracted from this model was then given as input to a bidirectional LSTM-based classification network known as BiLSTM. The performance of the model was evaluated using precision, recall, and f-score. It was also noted that uniform lighting and having a plain black background in the input images helped increase performance with fewer samples of data used. The proposed model gave an average classification accuracy of 83.36%. The model was also applied to two other datasets namely, the Cambridge hand gesture dataset and another ISL-based dataset giving accuracies of $99.34 \pm 0.66\%$ and 97%, respec-

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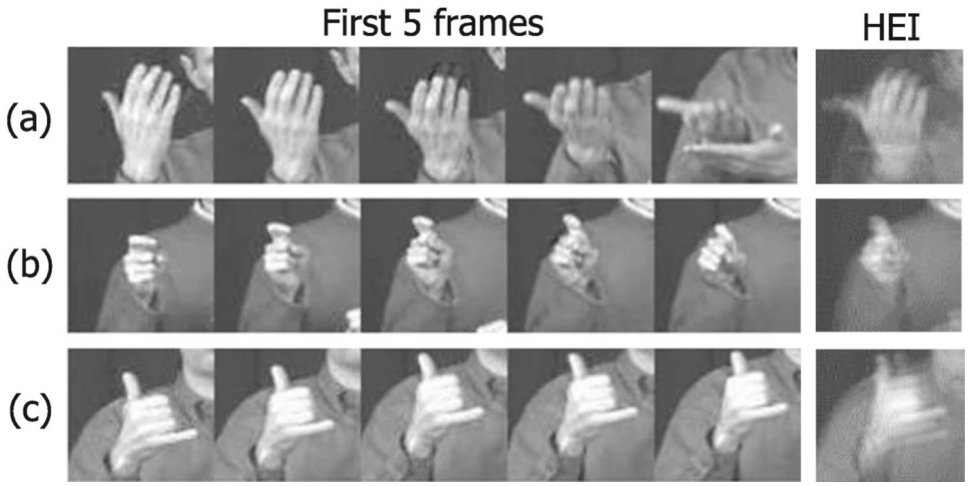


Fig. 4 Examples demonstrating how HEI is calculated by taking the mean of all hand regions. Examples are taken from [37]

tively. This was higher than that achieved by some other state-of-the-art models trained on the same datasets.

3DCNN was also used in [40] to extract important features from small spatio-temporal patches. These features were combined with the features extracted by a 2D convolutional recurring neural network (2DCRNN) from each frame. These features were used to train a neural network. The model was tested on the ArSL dataset. Results show that 3DCNN outperformed 2DCRNN with an accuracy of 99% (vs 92%).

Boháček and Hrúz presented a computationally low-cost solution while using the transformer model [41]. *Transformer* is a sequence-to-sequence (seq2seq) model that can transform a sequence (such as gestures) into another sequence (such as words). Like any seq2seq model, it is composed of an encoder and a decoder. The encoder is like a feature extraction module that compresses the input to a highly meaningful, fixed-length vector called a context vector. The decoder unwraps the context vector and tries to generate the desired output. Both the encoder and decoder can take variable-sized input, thus making them useful tools for sequence-to-sequence modeling. Additionally, what makes transformers special is the attention mechanism. Besides SLR, the attention mechanism has been used in NLP for example, for image captioning [42] and pattern recognition, for example for face recognition [43]. When processing a fragment f in a sequence, the attention mechanism helps to identify which other fragments in the sequence are important for processing f.

The study analyzes the ability of the transformer model to recognize isolated sign language based on body-pose representation [41]. The model proposed in this article is called the SPOTER - *s*ign *po*se-based *t*ransform*er*. The proposed pipeline begins with the extraction of landmarks using the vision API algorithm. A total of 5 head landmarks, 54 body landmarks, and 21 landmarks per hand were obtained. Data is also augmented. The spatial augmentations used were in-plane rotation, squeeze, perspective transformation, and sequential joint rotation. The normalization process is performed to prevent the model from learning irrelevant spatial features. The proposed architecture is a modified transformer mentioned by Vaswani et al [44]. The input to the model is the normalized 54 joint locations and ultimately 108-dimensional pose vector for every image. The transformer consists of 6 encoder and decoder layers, respectively, and 9 heads in each. The result shows that the SPOTER model delivered 63.18% and 43.78% accuracy for WLASL100 and WLASL300, respectively, which is a 3% and 1.6% improvement, respectively, from the previous state-of-the-art pose-based approach. Other models that performed better came with a higher computational cost. The LSA64 dataset gave an accuracy of 100% with the proposed model. 10 SPOTER and I3D models were trained for 10 different sizes of the training data and the SPOTER model outperformed I3D by giving an accuracy of 88.68% while the I3D model had an accuracy of 45.47%.

In [45] a state-of-the-art object detection method called single shot detector (SSD) for hand gesture recognition is used. Before proceeding further, *SSD* is briefly explained. Object detection algorithms generally pass a sliding window across the entire image and use an object classification algorithm to determine if the object is present in that window or not. These approaches are not suitable due to their dependence on the size of the sliding window and the aspect ratio. Other mainstream methods are based on two steps: identifying probable regions and detecting objects in those regions. SSD can find objects in the image using just one pass. It has a pre-trained classifier followed by convolutional layers that generate bounding boxes representing the locations of detected objects when an image is given as input. In this study, SSD is used to detect hands. The study proposed a

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complete pipeline for signed word recognition, and SSD is one of the components.

Besides SSD, models like 2DCNN, 3DCNN, and LSTM were also used. After the hands have been detected by SSD, the bounding boxes are given to a 2DCNN model that converts 2D images of hands into 3D key points. These 3D points are used to model a 3D hand skeleton that can be viewed from different angles. Heatmaps were also produced as outputs. 5 parallel 3DCNNs were used to extract features from all of these modalities such as frames of the hand region, the hand skeleton, etc. Finally, the features were fed to LTSM for training and classification. The proposed model achieved an accuracy of 99.80% on RKS-PERSIANSIGN and 91.12% on First-Person datasets. Using multi-modal data as input to the 3DCNNs, like the multi-view hand skeleton and heatmap features makes the model capable of outperforming many of the previously built models. Moreover, the use of a 3D hand skeleton for feature extraction improved the detection of hand gestures by allocating certain 3D key points on the hand for analysis. 3DCNN was also used for Pakistan sign language recognition with promising results [46].

In [47], a hand gesture recognition system is proposed that uses Hu moment to extract features from still images of hands and KNN for classification. This paper only proposed the model and no testing or validation was done. Another paper [48], used the CNN and RNN combination for feature extraction and sequence classification, respectively. Table 2 provides a summary of the studies covered in this subsection.

2.2.1 Static signs for words

This subsection discusses studies from the years 2019–2022 that focus on static signs for words. A static sign for a word means that the word is in the form of a static image rather than a gesture or a sequence. Figure 5 gives examples of words with static signs.

[51] proposed a simple CNN for ASL word recognition. Every word is in the form of a static image rather than a gesture or a sequence. The paper first extracts hands from the images using a skin color detection module and then only the images of hands are used to train a CNN. Since this study deals with just images, alphabets and numbers were also used to validate the model. The accuracy of static words was 90.04%.

Another study, that used static images to represent words instead of sequences is [52]. In this paper, openPose is used to extract important landmarks such as joints, fingers, etc. A total of 137 key points are extracted. 2D CNN pre-trained on ImageNet was also used separately to extract features. Relevant features from every frame were identified and fed to a transformer network consisting of a multi-head attention mechanism to classify the static signs. An accuracy of 74.7% was achieved for 100 words or classes. Another study [53] uses a convex hull to extract features and KNN for the classification of 5 static images of numbers from 1 to 5. The model yielded 65% accuracy.

In [54], the authors extend the dataset of a word-level SLR system using subtitled news videos. The paper argues that it is difficult to annotate sign data, however, there are a variety of news videos with both subtitles and a signer articulating those words through sign language. These can be used to augment the dataset. Note that this paper focuses on static signs for words. Using a sliding window and a pre-trained sign word recognition system, regions on the video frames representing signed words are extracted from news videos. Then a classifier is trained using both news signs and word signs to learn the align the features of these two signs that initially have different distributions because of different domains. Temporal attention is also employed to ignore the common gestures that occur in most of the words such as putting the hand down or raising for the next word, etc. The proposed model outperformed the recurrent convolutional neural networks (R-CNN) and I3D models. It remains unclear as to how were the

Finally, [55] proposed another pipeline based on the CNN model for the recognition of static signs from the ISL that include digits, alphabets, and routine words. The pipeline had four stages. (1) Image acquisition, where an RGB image consisting of static signs is acquired from the web camera in different conditions. (2) Prep-processing, where noise is removed, images are normalized and images are resized to 128×128. (3) Training, where the CNN classifier is trained on shuffled 80% of the dataset. (4) Testing, where the model is tested on the remaining 20% data. Testing was conducted using 50 different CNN models by playing around with different hyper-parameters such as the number of layers in CNN, the number of convolutional filters, and optimizers. Three different optimizers were tried: Adam, RMSProp, and SGD. The results showed that SGD surpassed RMSProp and Adam for both colored and grayscale images. These models were compared with other existing systems like KNN, SVM, ANN, and CNN. It was noted from the results that the proposed deep CNN succeeded all other ISL recognition systems with an accuracy of 99.90%.

2.3 Sentences

Sentence-level or continuous SLR is considered the most complex type of language recognition mode. In addition to the challenges of signed word recognition, challenges related to seq2seq modeling for converting signed words to spoken words also need to be addressed. Glosses segmentation or alignment, and reliably extracting gloss boundaries are some of the areas that studies in this mode of SLR focus on, and these tasks still remain an open research problem in SLR. Converting glosses to text can also be cumbersome.

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Table 2 A summary of studies on SLR that are based on word-level

References	Model/technique	Dataset	Best acc
[32]	VGG-16 (transfer) + RNN	ASL	32.63%
[33]	2DCNN (pre-trained) + CTC	RWTH-PHOENIX-weather 2014	–
[49]	T-UNet	CSI - 6 gestures	88.60%
[35]	Inflated 3DCNN (transfer on MS-ASL)	GSL	Up by 21 %
[36]	G-CNN+ Augment	ISL, ASL	96.43%, 100%
[47]	KNN. Hu feature extraction	–	–
[48]	CNN+RNN	RWTH-PHOENIX-weather 2014, SIGNUM	22.8% WER, 2.8% WER
[38]	Resnet (transfer) + attention + Augment, I3D	AUTSL	94.83%
[39]	VGG-16 (transfer) + LSTM	ISL, Cambridge	83.36%, 99.34%
[40]	2DCRNN, 3DCNN	ArSl	93.0% (3DCNN)
[41]	Transformer. Landmark extraction	WLASL100, WLASL300, LSA64	63.8%, 43.78%, 100%
[50]	SSD, CNN, SVD, LSTM	ASL	92.7%

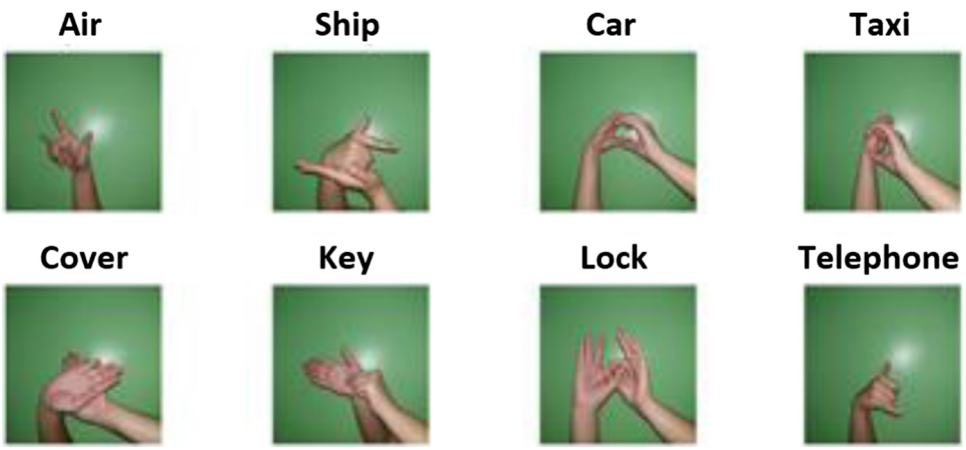


Fig. 5 Examples of static images representing words [51]

Glosses are sequences of transcribed spoken language words in the order in which they have been signed. The sequence of glosses can be different from the sequence of spoken words. Typically, sentence-level SLR consists of three phases [56] as shown in Fig. 6.

In the first phase, alignment of signed video with glosses. In the second phase, the glosses are recognized, and in the third stage, glosses are translated into spoken words. In some cases, the first two phases are combined as gloss recognition is used to assist in segmentation/alignment. In this section, we review recent studies (2019–2022) on continuous SLR and identify some of the challenges and future research directions in this area. We have categorized the studies based on their focus.

2.3.1 End-to-end SLR systems

End-to-end studies focused on the complete SLR pipeline as given in Fig. 6 i.e., from sign language to proper English sentences.

Zhu et al. [4] worked on sentence-level recognition where they addressed the issue of model fitting while recognizing continuous sign language by proposing an end-to-end SLR network known as a multi-scale temporal network (MSTnet). The model consists of three steps: frame-wise feature extraction, time-wise feature extraction, and training of the proposed multi-level CTC loss part. Frame-wise feature extraction consists of one main CNN and then 2 fully connected layers with 512 and 1024 neurons, respectively. The time-wise feature extraction is done through a proposed model of an MST block and transformers. The transformers use a self-attention mechanism instead of RNN that allows the model to have parallel training. The final part is employing CTC as a decoder. The length of the video sequence was shrunk and grown by 20% for temporal augmentation. A beam search algorithm with a beam width of 10 was employed in the CTC decoding stage. The frame-wise feature extraction was done using ResNet34 as the backbone and the MST block used for time-wise feature extraction had 1DCNNs along with 2DCNNs for feature fusion. Word Error Rate (WER) was used as the evaluation metric. The

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Fig. 6 High-level diagram of sentence-level SLR

proposed model showed a WER % of 21.4 for the test sets for the RWTH dataset which were the lowest numbers out of all the other state-of-the-art models. Moreover, the CSL dataset achieved a WER% of 0.7. The ablation studies conducted on the model showed that increasing the number of 1DCNNs to 4 gives the best results, i.e., the lowest WER % for the RWTH datasets. Moreover, two layers of fully connected layers produced optimal results. Furthermore, using transformers instead of BiLSTM was proved to be better as WER% was 1 and 0.3 percent less for validation and test sets of the RWTH dataset.

GoogleNet pre-trained on ImageNet was also used for continuous SLR [57]. This was used as a starting point to further train the network on the SLR data. At the end of the fully connected layers, a softmax layer was added to convert outputs into probabilities that could be used as posteriors for the hidden Markov model (HMM). The article employs three state-of-the-art datasets to evaluate the performance of the proposed model. The input to the model is a sign video that is passed as a sequence of images that the SLR model tries to recognize. For this reason, Bayes"^{TM} decision rule is followed. Moreover, the temporal variations in the input are modeled using HMM. The HMM part of the model is based on RASR, a state-of-the-art speech recognition system. Word Error Rate mechanism has been employed to evaluate the performance of the proposed mode. The three datasets result in the lowest WER% of 38.8, 30.0 and 7.4 for PHOENIX 2014, PHOENIX 2012, and SIGNUM, respectively. These results suggest a possible future study to improve the accuracies by using a different CNN model.

2.3.2 Hand detection

[58] worked on real-time sign language detection of five hand gestures using a user-defined dataset. Some challenges identified in the beginning were environmental disturbances, closure around areas of interest, and sign boundary detection. The proposed model utilized transfer learning from a pre-trained single-shot detector (SSD) mobilenetv2 model. MobileNet is CNN that explicitly focuses on image classification for mobile applications. Depth-wise separable convolution layers are used instead of the regular convolutional layers. SSD is used for object detection, and in this study is was used for detecting hands. It marks the area of interest during real-time with the help of bounding boxes using the labeling library. The detected images of the hands are fed to mobilenetv2. The highest accuracy achieved was 91.0% for the gesture 'hello' whereas the lowest was 82.4% for the gesture 'I Love You'.

In another study, hand movement was used to segment sequences into words [59]. Hand movement provides interesting cues for the location of word boundaries in sentence-level SLR. Analyzing hand movements require the detection of hands. For this purpose, the faster R-CNN technique was used. The main limitation of this study is consecutive signs with no visible pause makes it difficult to detect word boundary.

2.3.3 Alignment

Liao et al. [59] also attempt to solve the alignment problem in continuous SLR. Essentially, it consists of a CNN that extracts features while bi-LSTM learns the sequence. Before moving on to the classifiers, the sequence needs to be segmented into words. The movement of the hands and arms, such as pause, etc. were used as cues to predict the end of a signed word. The faster R-CNN technique was used for detecting hands. This not only helps in detecting the word boundaries, but also reduces the time complexity by only feeding the images of hands to the CNN classifier, instead of the complete frames. An accuracy of 89.8% and 86.9% were obtained on the DEVISIGN_D and SLR_Dataset, respectively.

Gao et al. [60] present another study that focuses on alignment between the video sequence and the sentence label. For this purpose, they introduce an RNN-Transducer that learns a mapping between the various components of the two sequences. A lexical prediction network is proposed much like a next-word-prediction model, that infers and uses contextual information to predict the next outputs. The model was evaluated on a public Chinese dataset consisting of signed sentences. The dataset has 100 sentences based on 178 words. Results show that the model is superior to other state-of-the-art models used with the same datasets.

2.3.4 Feature extraction

Spatial and Temporal Cues

A very interesting study was conducted by [61] in which spatial and temporal cues were extracted from signed sentences. The modules were called spatial multi-cue (SMC) and temporal multi-cue (TMC). The SMC module first decomposes frames into cues such as hands, faces, etc., and then spatial extracts features from these patches in addition to extracting

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features from the full frame. These features are passed on to the TMC. The TMC identifies unique features of each cue and also combines features of different cues at different time scales. The idea is to calculate temporal correlations within and between cues. All of this information is then passed on to BiLSTM encoders and CTC layers for sequence learning and inference. The model was tested on 3 datasets and it outperformed state-of-the-art in all cases.

The model SMC and TMC modules proposed by [61] were first used by [56] to develop a complete translation system that generates a meaningful spoken sentence from a stream of frames containing the sign language instead of just glosses. For sign-to-gloss recognition, a transformer t_1 was used based on SMC and TMC and was called STMC. In addition to using the cue-based transformer, another transformer t_2 was used that was trained on video and ground truth gloss annotations. It was reduced to only 2 encoder-decoder layers instead of 6, as originally proposed by [44]. The two transformers t_1 and t_2 were compared and the results show that previous claims that ground truth gloss annotations always produce an optimal solution were found to be false, as STMC or t_1 performed better. Once again, the SMC module decomposed a video in spatial features, and TMC was used for temporal pooling and to calculate temporal correlations within and between cues. It was noted that the transformer produced good-quality results despite having lower BLEU scores. However, due to the dataset being limited to PHOENIX-Weather only, the domain of the trained network is also limited to forecasting. Both spatial and temporal feature extraction modules were also proposed in another study [62]. Attention mechanism was also used to extract and focus on relevant features in both the spatial and temporal phases of the model. Frame-wise spatial features were extracted using ResNet and passed on to an attention module which was followed by the temporal module.

In [63] a hybrid deep-net model for ISL recognition is proposed. The model consisted of GoogleNet, a pre-trained CNN model, and a bidirectional LSTM (BiLSTM). The transfer learning approach was adopted where a pre-trained GoogleNet model was used to extract spatial and temporal patterns from hand gestures in the given video frames. This study also introduced a dataset that was created using real-time video frames for 13 commonly used agricultural terms. It was shown that average classification accuracy of 76.21% was achieved. To improve the accuracy a larger dataset is needed. Finally, in [64], weakly annotated signed sentences are used to train a fully convolutional network (FCN) that learns important spatial and temporal features.

Topological Features

Topological features relate to the anatomy of the hands such as finger tips, joints, distance or angle between fingers, finger joints, etc. In this section, we review some recent studies that used topological features for SLR. [65] proposed a 3D SLR system using color-coded topographical descriptors consisting of joint distances and the angles computed from joint locations. An ISL 3D dataset (ISL3D) was created using a 3D sign skeleton with 8 cameras to capture 98% of the joints in each sign. For this purpose, the skeleton was made up of 57 upper body joints: 4 on the head, 12 on the face, 1 on the chest, 2 on each shoulder, 2 on the arms and 18 each for the left and right hand. A 2 stream CNN having parallel convolutional layers was used that were combined just before the dense layer. The proposed model takes 2 types of color-coded images as input namely, the joint distance topographic descriptor (JDTD) and the joint angle topographical descriptor (JATD). Since most of the datasets are limited to hand movement recognition and not fingers, this caused issues in performance and accuracy. So this article also focused on developing a dataset of ISL using an upper-body human joint 3D model. The spatial and temporal movements of joints in 3D space were captured to create JDTD and JATD color-coded maps which were then able to highlight the movements of the fingers. Moreover, 2 other existing skeletal datasets were also used on the proposed 2-stream CNN model namely, HDM05 and CMU. The proposed model (JDTD + JATD + CNN) achieved an average accuracy of 92%. Some hindrances noted in the proposed model included difficulty to cater for some complex signs which included movements of both hands.

Furthermore, another paper[66] focuses on SLR based on key point detection to produce captions for the speaker trying to communicate their message. The article initially works on detecting ASL alphabets and then aims at identifying the gesture-to-sentence translation using a CNN model. The study begins with data pre-processing where PCA was performed to reduce the size of images. ASL alphabet recognition was done through mediapipe which collected 21 landmarks. The classification of these gestures was done using the KNN, decision tree and random forest classifiers. The main issue identified in this research is the missing key points in some frames of the video. This was resolved using the mean value of the previous and upcoming frames, however, a better approach can be used. Sentences of various lengths were normalized by padding with 0. The gesture-to-sentence translation was performed using a transformer that had an encoder followed by a multi-head self-attention module. The encoder also contains a feed-forward network and a layer to normalize the output of the feed-forward network. The input to this encoder was 18 landmarks. Simultaneously, positional encodings are fed to a masked self-attention layer which is followed by another normalization layer. The output of this layer is passed to the encoder-decoder module. This module tries to grasp the mapping of the input to the target sequences. The output of this module is fed to the feed-forward network which is followed by a softmax layer.

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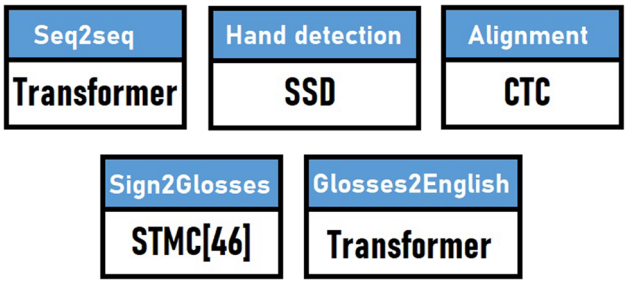


Fig. 7 State-of-the-art techniques for different task /stages of sentence-level SLR

A limitation of this study was that the proposed model was not tested and hence does not provide any results.

In addition to using mediapipe for extracting key points, high resolution net (HRNet) [67] has also been used in [62]. HRNet is pre-trained on 57K images labeled with 17 key points and is able to estimate the 17 key points of human body. In the proposed study, these key points were extracted frame-wise. Then the spatial features of these key points were extracted using ResNet. Attention mechanism was also used to focus on the relevant features.

2.3.5 Discussion

An observation made during the review was that every study is disconnected. There is no series of studies that gradually improves any one particular aspect or method for sentence-level sign recognition. As a result, there is no conclusive opinion on the most valuable approach or method. This indirectly hinders the overall research in this area. Nevertheless, we do identify some of the state-of-the-art models and techniques that are being employed for different phases (see Table 6) of sentence-level SLR. These are shown in Fig. 7.

Having said that, transformers are the best options for sentence-level SLR since they provide a tailor-made framework for seq2seq tasks in an end-to-end manner. Signed sentence language recognition is a seq2seq mapping problem as it aims to convert a signed sequence of words into a sequence of spoken ones. The use of transformers is further supported by [56]. For sign-to-glosses recognition, the well-known approach was to train a classifier on video and ground truth gloss annotations. However, in [56], it was shown that ground truth gloss annotations did not produce the optimal results and the transformer-based STMC model performed better.

Furthermore, another issue with most of the studies is that they were validated on very small data. There is a need to integrate a thorough objective evaluation methodology into image restoration studies. Finally, some studies employed transfer learning models, but that did not produce good results.

2.4 Sign language recognition using devices other than cameras

Wearable devices data gloves, wristwatches or armbands, etc. are also used as input for SLR. Methods based on such wearable devices have produced good results. The studies that employ wearable devices to record signs are discussed in this section. Data gloves can be intrusive, but wristwatches or armbands can be more convenient to use.

In [68], wearable gloves were used for both sign word and sentence recognition. The glove consists of triboelectric sensors neatly placed within the glove textile. This sensor is extremely sensitive to the physical movement of the fingers and hands and can capture or record different motions. The glove used in the study had 15 channels. For a word and sentence, 200 and 800 points were collected, respectively. Two models were proposed, one that would process the sentences as a whole in addition to words. This model is CNN and for words, it would take as input all 200 data points (800 for sentences) from 15 channels stacked on top of one another as shown in Fig. 8. The other model would segment the sequence into sign words. Using a sliding window, the sign sentences were split into fragments and each fragment was recognized. Then these fragments were re-ordered to match the training sentence which was then sent to the first CNN model that would recognize the sentence. For validation, 50 words and 20 sentences were correctly recognized. Unseen sentences formed by only rearranging the words in the vocabulary were recognized with an accuracy of 86.67%.

In [69], another wearable device was used for ASL word and sentence recognition. An armband device was used called Myo armband 1 can be connected to a mobile phone through a Bluetooth connection. The device provided readings for the accelerometer, gyroscope, orientation, and 8-channel EMG signals. Each sample, therefore, consists of 17 channels in total. Like many other studies discussed in this paper, this paper also uses CNN to extract important features from each channel which are then fed to a bidirectional Long Short Term Memory (LSTM) that learns the temporal information. The paper further extends the model to sentences and propose a connectionist temporal classification objective function to map sign with word and achieve sentence-level SLR. Results show that MyoSign achieves an accuracy of 93.1% at the sentence level.

Microsoft's Kinect has also been used for SLR by [70]. This study is interesting for two reasons. The first one is the usage of a Kinect device. Kinect sensors have not been used a lot for SLR in the past, but their utilization is steadily increasing [71]. This device has three input components making it different from simple camera-based SLR: a depth sensor, a multi-array microphone and a video camera. The second interesting aspect of this study is the introduction to Brazilian sign language (BSL), which according to the authors is one of

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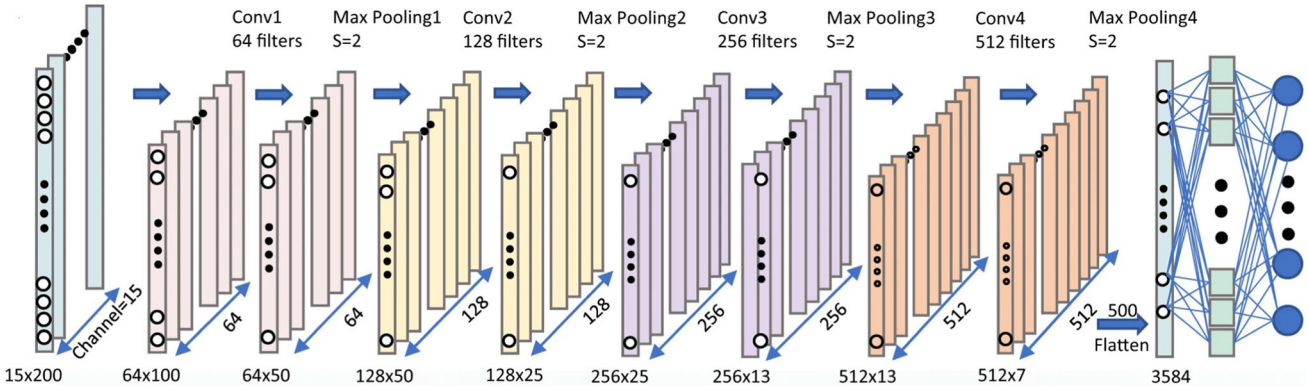


Fig. 8 200 data points of a word from all 15 channels are stacked together and fed to the CNN for training and recognition [68]

the few sign languages that follow a linguistic structure similar to spoken languages. Thus expressions like 'strong rain' and 'weak rain' have similar signs since they share a similar linguistic unit. This study employed 2 multi-stream CNN architectures to handle the 3 input channels of the Kinect sensor. [71] also used Kinect sensor and LSTM as the classification model.

In [72], leap motion sensors were used for ISL. The sensor has two infrared cameras and three infrared LEDs. For each signed sentence, 12 features were extracted using CNN for each dimension (x,y,z). These features were passed onto a proposed long short-term memory (LSTM) model. The model was tested on 942 sentences formed using 35 signed words producing an accuracy of 72.3%. The leap motion device also provided information on the joints of fingers and wrists, but this was discarded. A leap motion sensor was also used in a very interesting study by [73] who developed a game called whack-a-mole, that was controlled using letter-level sign language. The game basically has holes labeled with English alphabets, and a mole would appear on one of the holes. The user would need to whack the mole by making the appropriate sign for the hole's letter. For this application, a leap motion sensor was used. The proposed model detected the changes in hand movement from one letter to another and use the changes in hand shape and position to predict the letter. For prediction, this study used LSTM. Before data was sent to the LSTM, features were extracted such as the movement of the hand, the relation between fingertips and between palm and fingertips, etc. The 26 English alphabets were recognized with an accuracy of 99.4%. Leap motion controllers have also been used for SLR such as in [74]. A leap motion controller or a sensor detects and tracks hands with its built-in cameras and infrared sensors. The study was based on the 26 alphabets and 10 digits of ASL.

Distortions and changes in WiFi signals were used to recognize actions in [49]. This study covers semantic segmentation at image level and video-based recognition. We only focus on the later. Although this study is related to gesture recognition and not SLR, we have included this study primarily because of the mode of input. Instead of precisely video, channel state information (CSI) is used as input. Every $i-th$ CSI sample frame is either categorized an action class, or background. This is done by temporal Unet or T-Unet model that identifies actions at the sample-level. T-Unet model contains three main components namely, the 'down"^{TM}, the 'up' and the 'shortcut links'. The down block reduces the size of the feature map and has two convolutional layers and one max pooling layer. The up layer increases the feature map and the input of dimension 48–256 is up-sampled by a deconvolutional layer to 96–128. Lastly, the shortcut link combines features from different layers to increase the channel. The CSI distortion dataset was used that consists of 1116 WiFi series for 6 different gesture actions. The T-Unet model was evaluated using the average precision (AP) metrics.

Surface Electromyography is a technique that can be used to record the activation of muscles. This is done by placing non-invasive electrodes near the muscle group. It is mainly used for analyzing muscle movement for the rehabilitation of individuals however, it has also been used for SLR [75]. In this study, an attempt was made to recognize 12 signs from Columbian sign language by placing electrodes on four forearm muscles. The electromyography signals obtained from these 4 channels were used to train the SVM classifier. Additionally, accelerometers were also used as a mode of input that were placed on the wrists. Different combinations of the readings from the 4 muscles and wrists were tried and the best results were obtained when all inputs were utilized.

2.5 Summary

In this section, we present a summary of all three units of SLR: alphabets, words, and sentences. The summary is in tabular form and given in Table 3. The table outlines the type of dataset, learning technologies, and challenges for each of the 3 types of SLR modes covered in this review.

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Table 3 High-level summarization of the research on alphabet-level, word-level, and sentence-level SLR

Type	Dataset type	Learning technology	Challenges
Letter or static word	Images	Similar to written character recognition: Any ML or DL	Uncontrolled environment
Word	Video or gesture (sequence of images for one word)	3DCNN, Inflated CNN, RNN, LSTM, transformers, transfer learning, SSD	Hand detection, many-to-one sequence classification, dynamic background
Sentences	Sequences of gestures	3DCNN, RNN, LSTM, transformers transfer learning, CTC, SSD	seq2seq classification, gloss segmentation/sign to glosses mapping, gloss to spoken sentence translation, hand detection, segmentation via hand movement, dynamic background

3 Dataset

This section provides a bird-eye view of various data sets, listed study-wise and used by researchers for SLR studies conducted during the period 2019–2022. These datasets are given in Table 4, along with a brief description. The purpose of this table is two folds. Firstly, to provide future researchers with a list of datasets that are currently being used and allow them to experiment with them. Secondly, by listing these datasets study-wise, we provide at least one baseline model for each of the datasets for future researchers to compare and improve upon.

4 Open research questions and future directions

A lot of work has been done on SLR in the recent past, however, there are still so many challenges, open research questions, and future directions that still need to be worked on. In this section, we provide a comprehensive list of open research questions for future researchers.

Segmenting continuous signs from a video and being able to reliably map the different sequences of sign videos to class labels or words is yet another open research question. Different techniques have been proposed such as using hand movements as cues or using a sliding window to get video fragments and then detect those fragments using a sign word recognition system, but all of these approaches have their shortcomings. Transformers have been used successfully for attention-based classification, these can also be employed for SLR. Having said that, transformers require a huge amount of training data. Also, other cues can be used to differentiate between words in a signed sentence such as facial expressions. In [80], facial landmarks were used, but to detect sign and their proposed model produced good results. Similarly, incorporating facial expressions in the segmentation module is another research problem. Limiting the recognition of signs to just the movement and shape of the hands can never produce a perfect score. The motion of the face and neck during sign language, no doubt a challenge on its own, can provide useful cues about the sign and its word boundaries.

Recently, a lot of transformers and visual transformer models have been proposed such as ViT, BERT, BART, etc. some of which are also available as pre-trained models. Models like BERT which have been used very successfully for natural language processing (NLP), can also be utilized for sign recognition. In fact, in one study [81], BERT models were used, one for each input modality for their attention mechanism. Then a synchronization module was introduced within each BERT to synchronize the attention mechanism across different modalities. Similarly, there are many other applications of BERT, etc. within NLP such as next word-prediction that can also be applied to SLR. In this scenario, a fragment of a sign sequence could be used to predict the remaining sequence, which could ultimately be used to recognize the entire sentence.

The pre-trained visual transformers can also be employed for sign recognition. However, further study is still needed to identify which pre-trained transformer model is best suited for this problem and how can they be adapted to produce even better results. Employing a pre-trained transformer model for sign recognition is a trending research problem. Furthermore, more efficient and less computational-power-hungry transformers have been proposed such as reformer and switch transformers for other domains. An interesting research question could be: can these also be used for sign recognition?

The normal practice in SLR is to use a single device for capturing signs such as a camera, leap motion sensor, gloves, etc. Multi-modal input is an interesting area that deals with solving problems using more than 1 channel. Using a camera along with a leap motion sensor to recognize continuous sign languages more accurately is an open research area both in terms of synchronized data collection and actual prediction of signs using state-of-the-art machine/deep learning techniques. In [49], distortions and changes in WiFi signals were used to recognize 4 gestures. Using such an input mode for SLR is a novel idea.

Every language of the world presents its challenges and limitation. During this review, it was observed that several

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Table 4 Details of various data sets used for SLR

References	Dataset
[11]	The system is capable of recognizing 125 words of everyday use including alphabets. Some of the ones mentioned in the article include “A”, “E”, “Family”, “I am”, “Goodbye”, “Hello”, “Help”, “Want”, “You”, “Language”, “Elevator”, “No”, “Clean”
[28]	Alphabets belonging to ASL
[20]	The ASL MNIST dataset was used excluding letters J and Z
[24]	The dataset consists of 54,049 images of Arabic sign language alphabets performed by more than 40 people for 32 standard Arabic signs and alphabets. There are around 1500 images per class, and each class represents a different meaning through its hand gesture or sign
[25]	ASL Alphabets A to H
[27]	A wide range of vocabulary was used from different languages including Hindi, Arabic, Chinese, and English. Data was also collected from special input devices including digital still cameras, video cameras, cyber gloves, and sensor gloves. The database not only covers alphabets, words, and small phrases
[23]	The first dataset contains 52,000 images of the ASL. The second dataset includes 62,400 images (github.com/evilport2/sign-language). The third dataset was developed specifically for this study which contains 104,000 images of 26 English alphabet
[29]	The dataset consisted of 24 ASL alphabets excluding “J” and “Z” that were collected from 4 individuals and 30 samples for each alphabet were gained. The distance at which the images were captured was 1.5 feet to 3 feet. The backgrounds of the dataset images were plain with only the hand in the frame
[30]	The dataset contains 6633 static images of the PSL alphabets. There is an average of 184 samples per alphabet. The images were captured under varying conditions including the position of the hand, lighting, scaling, and angles. The distance had a variation of 1 foot, 1.5 feet, and 2.0 feet. The background and clothing were not uniform
[32]	A large-scale Word-Level ASL (WLASL) video dataset, containing more than 2000 words performed by over 100 signers was used. It is noted that each sign is performed by at least 3 signers. The data was collected through educational sign language websites like ASLU and ASL-LEX. Youtube video tutorials were also used however, it was ensured that only those videos were selected that had gloss descriptors given in the video title. A total of 68,129 videos of 20,863 ASL glosses from 20 different websites were collected for the study
[33]	For the dataset 9 signers were used to create a dataset of 5672 sequences, while textual annotation contained 1231 words with only 410 singletons. This dataset contained both full frames and cropped-out a dominant hand
[49]	CSI dataset was used. Six gesture actions, i.e, “hand up”, “hand down”, “hand left”, “hand right”, “hand circle”, and “hand cross” are taken into account. The training set consists of 1116 WiFi series, with 6 categories of gestures performed by a signer in 16 indoor locations. The total size is 278
[56]	Two datasets were used, one was the PHOENIX-Weather 2014T which is taken from a German weather forecasting channel. 9 different signers volunteered to perform gloss-level annotations with a vocabulary of 1,066 different signs that were then translated to German spoken language with a vocabulary of 2,887 different words. The other is ASLG-PC12 which is English data from Project Gutenberg
[35]	As mentioned above, MS-ASL and SIGNUM were used. MS-ASL was collected from 22,063 Youtube videos, while SIGNUM was performed by 25 different signers, each signer performing the same sign 3 times. This resulted in a total of 11,250 recordings of isolated GSL signs (each of the 25 signers performed each of the 450 signs once)
[55]	The dataset included an array of 35000 RGB images with 350 images for each static sign. The dataset includes 23 English alphabets, 10 digits and 64 common words that were used in routine like ”water”, ”stand”, “fever”, ”hand”, etc. The dataset was composed of images of multiple sizes, colors, and shot under various environmental conditions
[36]	The dataset is a collection of 2150 images gathered by 50 ISL signers with 43 classes. The dataset has two categories where category 1 is Fingerspelling alphabets and category 2 is Isolated words. The second dataset used is Jochec-Triesch”s dataset of ASL which contains 3000 samples with 10 various classes with both light and dark backgrounds
[38]	AUTSL BosphorusSign 22k
[41]	The study used the WLASL dataset and the LSA64 dataset. WLASL contains signs gathered by native American SL interpreters or signers. Furthermore, the LSA64 dataset contains 3200 videos from Argentinian SL of 64 various glosses. These include both verbs and nouns performed by 10 non-expert individuals
[58]	The dataset over 2000 images consisting of 5 symbols, i.e.,Hello, Yes, No, I Love You and Thank You
[39]	Collection of three datasets: 1) Cambridge hand gesture dataset: This dataset includes 900 video samples of nine hand gesture classes. 2) Alternative ISL word dataset for emergencies: Includes words like ”accident“ ”call“, ”cut“, ”doctor“, ”help“, etc. 3) The proposed novel dataset with emergency words used by deaf COVID-19 patients. This dataset includes videos of 17 hand gestures corresponding to the ISL words ”breath“, ”call“, ”ough“, ”distance“, ”difficult“, ”doctor“, ”help“, etc

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Table 4 continued

References	Dataset
[65]	Introduced a dataset created by the 3D skeleton model consisted of 50,000 3D sign videos of ISL divided into 500 classes. Additionally, 3 existing datasets: HDM05 action dataset, 1500 actions from 23 classes were used, CMU action dataset, 700 actions from 70 classes and NTU RGB ""D human action dataset, skeletal data of 2400 samples from 20 subjects
[45]	Collection of three datasets: 1) RKS-PERSIANSIGN: This consists of 10 000 videos of 100 Persian signs using 10 contributors in 10 different backgrounds. 2)First person dataset: Has more than 100K images and depth frames for 26 different objects in several hand configurations. 3)NYU hand pose dataset: Has 81,009 images and depth of 36 hand joints
[63]	The proposed dataset includes RGB videos of 13 dynamic hand gestures for the words "animal", "banana", "coconut", "cut", "fire", "help", "insect", "monkey", "rat", "rice", "snake", "tiger" and "wheat". The hand gestures were captured according to the styles and movements specified in the ISL dictionary published by Ramakrishna Mission Vivekananda University, Coimbatore, Tamilnadu, India (FDMSE, 2016)
[76]	The proposed method is used on two main datasets; PHOENIX14 and CSL. Data was collected from German weather forecasting channels and contains 6841 sentences with 1295 different glosses. The dataset is then divided into various categories which are as follows; 5672 training sentences, 540 development (Dev) sentences, and 629 test sentences for the multi-signer setup. 50 signers performed each sentence 5 times
[77]	A new GLS dataset is introduced. 7 signers are used to collect the data. The dataset consists of 10,295 sentence instances, 331 unique sentences, 40,785 gloss instances, 310 unique glosses for the Greek Language
[78]	Most widely used datasets are ASL,CSL and PHOENIX. PHOENIX is the GSL
[66]	This study used ASL alphabets. The dataset had 1728 images and was signed by only one person. It contained 720 unique images and the rest were augmented using flip, crop, and rotation techniques
[66]	How2Sign dataset contains 80-hour-long instructional sign language videos performed by 11 signers out of which 5 could hear, 4 were completely deaf and 2 had issues with hearing. 4 out of 5 people who could hear were professional ASL interpreters and 1 was ASL fluent
[4]	Used the RWTH dataset that was recorded by public weather radio and television stations in Germany. 9 signers with clean backgrounds recorded 6481 sentences using 1232 words
[4]	This study also used the CSL dataset recorded using a Kinect camera. It consists of 100 routine Chinese phrases signed by 50 signers 5 times. The vocabulary size is 178. The CSL dataset was recorded with people wearing different clothes and at varying speeds and ranges of motion
[57]	Collection of three datasets: RWTH-PHOENIX-Weather 2012, RWTH-PHOENIX-Weather Multi Signer 2014 and SIGNUM single signer
[79]	60 thousand images of common poses and 25 thousand images of yoga poses were used. 33 points on the human body were considered a landmark

languages have been considered for SLR such as English, Arabic, Hindi, Turkish, etc. However, many other under-resourced languages have no (such as Sindhi) or very limited (such as Urdu) research studies. One main reason for this is the limited size or unavailability of these datasets. As a result, it becomes difficult to conduct formal studies to automate the SLR process for these languages. Thus, one interesting problem that can be worked upon in SLR is the development of annotated corpus in other less-resourced languages and also the development of baseline models for these datasets. Furthermore, another open challenge is to combine datasets of different languages and develop a universal sign recognition system that can not only detect the language of the signer, i.e., whether it is ASL, ISL, etc., but also the words and sentences in that language.

Mediapipe [82] has been used a lot in machine learning for processing time-series data or streaming data such as videos and gestures. However, mediapipe performance is limited with longer sequential data resulting in poor learning efficiency [83]. For more tool-specific research, it would be interesting to optimize mediapipe using evolutionary algorithms. Furthermore, HRNet has also been used for detecting landmarks [62]. HRNet is pre-trained on 17 key-points-annotated images and is able to estimate the 17 key points of human body. For future studies, the performance of HRNet can be compared with mediapipe. HRNet itself has not been validated. At present, it has only been validated indirectly through the overall performance of the SLR system. It would be worthwhile to investigate the performance of HRNet at key point level with metrics such as number of key points detected correctly per frame, etc. Furthermore, instead of HRNet, other pose estimation models that identify key points can be used.

Another research idea presented in [84] and not directly related to SLR is to convert sign language to speech using a speech recognition system, and then the speech is given to a robot that performs the sign. Eventually, this system can be used for question/answering between hearing-impaired and normal individuals where a user could type the text, question, or answer, and a robot performs the gestures.

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Some studies have used faster R-CNN for detecting hands including [59]. For future work, other state-of-the-art techniques like YOLO can be employed.

Another observation made during the review was that every study is disconnected. There is no series of studies that gradually improves any one particular aspect or method for SLR. As a result, there is no conclusive opinion on the most valuable sequence segmentation approach, features extraction method, or classifiers. This indirectly hinders the overall research in this area.

5 Conclusion

This paper presents a detailed and comprehensive survey of the state-of-the-art SLR techniques developed from the years 2019–2022. The review delves deeper into different SLRs based on linguistic units: alphabets, words, and sentences, and presents the various models proposed by many researchers to detect sign language. The models vary from CNN, hybrid CNN, and VGG to transformers. Several languages in the world have their sign language. In this survey, we covered sign recognition techniques employed for various languages such as ASL, Chinese SL, Argentinian SL, Turkish SL, German SL, and Indian SL. The work presented in this paper lays the ground for future research into SLR techniques as it not only encompasses high-level summaries of the previous work, but also the datasets that were used. Thus, providing the resources in one place that are required for conducting fruitful research.

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